

WILLIAMS LAKE, BC, Canada

WMO: 711040

Lat: 52.1862N

Lon: 122.0656W

Elev: 3083

StdP: 13.13

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-15.8	-8.9	-23.7	1.7	-15.1	-16.4	2.5	-8.4	29.4	33.4	26.1	33.4	3.1	140	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	83.0	59.1	79.4	57.9	76.0	56.5	60.7	78.1	59.1	75.8	57.7	73.0	6.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.8	71.7	62.6	52.8	66.4	61.6	51.1	62.3	60.7	28.2	78.2	27.1	76.0	26.1	73.2	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.2	18.8	16.6	DB	-23.6	88.3	8.4	3.1	-29.6	90.5	-34.5	92.4	-39.2	94.1	-45.3	96.4	(4)
(5)			WB	-24.0	63.5	8.3	1.9	-29.9	64.9	-34.8	66.0	-39.4	67.0	-45.4	68.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	20.6	23.5	32.0	40.4	49.6	55.8	61.1	59.9	51.5	40.2	28.7	20.2
	DBStd	17.20	13.93	11.83	9.96	6.65	7.06	5.83	6.02	6.00	6.99	7.33	10.61	12.28
	HDD50	4559	910	742	559	296	96	11	1	3	64	314	640	923
	HDD65	9041	1375	1162	1023	737	479	283	151	177	407	769	1090	1388
	CDD50	1052	0	0	0	9	83	184	346	311	109	10	0	0
	CDD65	62	0	0	0	0	1	6	32	21	2	0	0	0
	CDH74	1155	0	0	0	0	39	137	516	394	68	1	0	0
	CDH80	256	0	0	0	0	4	28	131	84	9	0	0	0
Wind	WSAvg	6.1	6.8	6.4	7.2	6.7	5.6	5.2	4.6	4.4	5.1	6.7	7.4	7.3
Precipitation	PrecAvg	17.8	1.6	1.0	0.8	0.9	1.5	1.9	2.1	1.8	1.5	1.4	1.5	1.7
	PrecMax	26.4	4.8	2.2	2.6	2.6	3.1	4.9	7.9	5.0	3.8	4.0	3.7	3.4
	PrecMin	10.2	0.3	0.3	0.2	0.1	0.3	0.6	0.2	0.3	0.2	0.2	0.2	0.3
	PrecStd	3.9	1.1	0.5	0.5	0.6	0.8	1.1	1.4	1.2	0.9	0.9	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	47.2	57.9	69.5	78.3	83.7	88.3	86.9	80.5	66.2	52.6	43.0
		MCWB	38.2	38.1	43.5	49.9	54.6	58.9	61.2	59.8	56.8	51.5	44.0	37.3
	2%	DB	40.7	42.8	51.9	62.1	73.3	78.0	83.6	82.1	75.2	60.0	48.1	39.4
		MCWB	36.1	35.5	40.5	46.7	52.7	56.4	59.5	58.9	55.7	48.7	42.2	34.4
	5%	DB	38.3	40.1	48.2	57.4	68.8	73.1	79.9	78.5	70.5	55.5	44.3	36.9
		MCWB	34.5	34.2	38.4	44.0	51.1	54.5	58.3	58.2	54.5	46.5	38.7	32.9
	10%	DB	35.9	37.5	44.7	53.1	64.3	69.0	75.9	74.7	65.5	51.9	40.7	34.3
		MCWB	32.5	32.7	36.4	41.5	49.4	53.2	57.0	56.8	52.4	44.0	36.8	31.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	39.3	44.6	50.8	56.4	60.5	62.9	63.1	59.0	53.0	45.6	38.3
		MCDB	42.9	45.1	56.6	66.7	74.1	77.9	83.1	79.1	76.2	63.4	51.0	42.1
	2%	WB	36.6	36.7	41.4	47.8	53.9	58.1	60.9	60.8	57.1	50.0	42.6	35.1
		MCDB	40.0	41.5	50.5	60.4	69.5	74.1	78.6	78.1	72.0	58.2	47.7	38.3
	5%	WB	34.7	34.7	39.2	45.0	52.2	56.0	59.3	59.0	55.1	47.4	39.5	33.3
		MCDB	37.8	39.2	46.5	55.7	66.1	69.8	76.2	75.1	68.3	54.7	43.1	36.5
	10%	WB	32.9	32.9	37.1	42.5	50.4	54.2	57.9	57.5	53.1	44.7	37.0	31.5
		MCDB	35.5	37.0	43.8	51.9	62.0	66.7	73.3	72.4	64.3	50.6	40.5	34.0
Mean Daily Temperature Range	5% DB	MDBR	12.4	15.8	17.3	19.8	21.4	20.3	21.9	23.3	21.7	16.5	11.6	11.3
		MCDBR	11.3	15.8	20.1	26.7	29.4	28.3	29.5	30.8	30.5	22.5	13.4	10.3
		MCWBR	8.2	10.4	11.9	14.8	13.6	12.0	11.4	12.7	14.7	13.3	9.2	7.9
	5% WB	MCDBR	10.8	14.7	19.0	24.2	25.7	24.4	26.1	27.6	27.5	21.0	12.5	10.4
		MCWBR	8.3	10.2	11.9	13.8	12.5	10.9	10.8	11.8	13.5	12.9	9.4	8.6
Clear-Sky Solar Irradiance	taub	0.258	0.266	0.288	0.313	0.333	0.334	0.325	0.327	0.300	0.286	0.270	0.259	
	taud	2.414	2.442	2.491	2.467	2.453	2.475	2.539	2.523	2.603	2.612	2.528	2.362	
	Ebn at Noon	246	277	292	294	292	292	292	286	285	268	241	221	
	Edh at Noon	17	23	27	31	33	33	30	29	24	19	16	16	
All-Sky Solar Radiation	RadAvg	285	594	1016	1440	1687	1803	1896	1613	1111	642	300	213	
	RadStd	24	55	71	103	119	148	176	75	106	58	30	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page